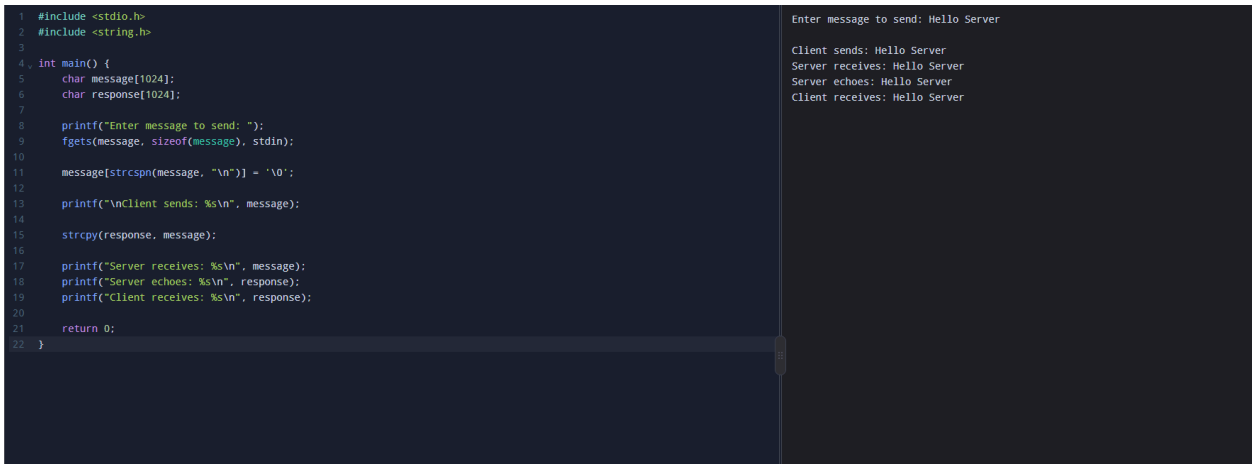
KEVIN INFANT P A - 241001116

12) A) Implement echo client server using TCP/UDP Sockets

Server Program



```c
#include <stdio.h>
#include <string.h>

int main() {
    char message[1024];
    char response[1024];

    printf("Enter message to send: ");
    fgets(message, sizeof(message), stdin);

    message[strcspn(message, "\n")] = '\0';

    printf("\nClient sends: %s\n", message);

    strcpy(response, message);

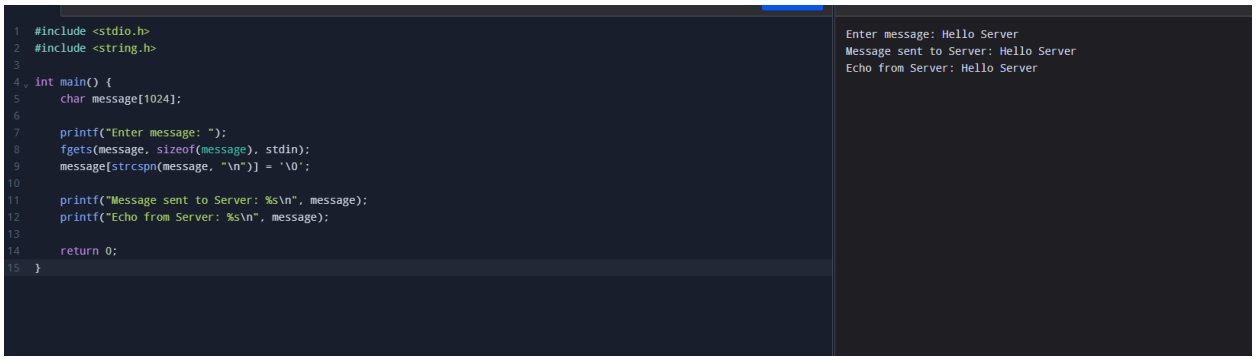
    printf("Server receives: %s\n", message);
    printf("Server echoes: %s\n", response);
    printf("Client receives: %s\n", response);

    return 0;
}
```

```
Enter message to send: Hello Server

Client sends: Hello Server
Server receives: Hello Server
Server echoes: Hello Server
Client receives: Hello Server
```

Client Program



```c
#include <stdio.h>
#include <string.h>

int main() {
    char message[1024];

    printf("Enter message: ");
    fgets(message, sizeof(message), stdin);
    message[strcspn(message, "\n")] = '\0';

    printf("Message sent to Server: %s\n", message);
    printf("Echo from Server: %s\n", message);

    return 0;
}
```

```
Enter message: Hello Server
Message sent to Server: Hello Server
Echo from Server: Hello Server
```

12) b) Implement chat client server using TCP/UDP sockets.

Server Program:

```c
#include <stdio.h>
#include <string.h>

int main() {
    char clientMessage[1024];
    char serverMessage[1024];

    printf("Waiting for client...\n");

    printf("Client: ");
    fgets(clientMessage, sizeof(clientMessage), stdin);
    clientMessage[strcspn(clientMessage, "\n")] = '\0';

    if (strcmp(clientMessage, "exit") == 0) {
        return 0;
    }

    printf("Server: ");
    fgets(serverMessage, sizeof(serverMessage), stdin);
    serverMessage[strcspn(serverMessage, "\n")] = '\0';

    printf("Client: %s\n", clientMessage);
    printf("Server reply: %s\n", serverMessage);

    return 0;
}
```

```
Waiting for client...
Client: hello from client
Server: hello from server
Client: hello from client
Server reply: hello from server
```

Client Program :

```c
#include <stdio.h>
#include <string.h>

int main() {
    char clientMessage[1024];
    char serverMessage[1024];

    printf("Client: ");
    fgets(clientMessage, sizeof(clientMessage), stdin);
    clientMessage[strcspn(clientMessage, "\n")] = '\0';

    if (strcmp(clientMessage, "exit") == 0) {
        printf("Chat ended.\n");
        return 0;
    }

    printf("Server: ");
    fgets(serverMessage, sizeof(serverMessage), stdin);
    serverMessage[strcspn(serverMessage, "\n")] = '\0';

    printf("Client sent: %s\n", clientMessage);
    printf("Server replied: %s\n", serverMessage);

    return 0;
}
```

```
Client: hello from client
Server: hello from server
Client sent: hello from client
Server replied: hello from server
```